ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MECHATRONICS

PAPER 1 · PRACTICE H · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Mechatronics paper 1 · practice h.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Accept other correct answers, sensible project decisions and neat, recognisable sketches, tables and circuits.

QUESTION 1

1.1 The design does the job it was made for, in the place where it will be used, for the people who will use it ✓, reliably and safely for its expected life ✓. (2)

1.2 Any FOUR sensible items with measurable values ✓✓✓✓, e.g. keeps soil moisture between 30% and 60%; switches the fan on above 30 °C; runs only on solar power with one day of battery backup; waters an area of 50 m²; water tank of at least 1 000 litres; total cost below R25 000; works unattended for 6 weeks. (4)

1.3 Water is scarce and the school pays for it, so the system must not waste it ✓; the solar supply is limited, so low-energy pumps, fans and controllers keep the system working and cost less ✓. (2)

1.4 Any TWO ✓✓: freehand or CAD sketches; 3D models or renders; a small working model / prototype; a presentation with drawings and costs; a flowchart or block diagram. (2)

[10]

QUESTION 2

2.1 Inputs: soil-moisture sensor ✓, thermostat / temperature sensor and float switch ✓; controller (PLC or microcontroller) ✓; outputs: pump ✓, fan and warning lamp ✓; power: solar panel, charge controller and battery feeding the system ✓ (arrows showing the direction of signals and power) (6)

2.2 The sensor measures the soil moisture (feedback) ✓; the controller compares the measurement with the set point ✓; if the soil is too dry, it switches the pump on ✓; the water raises the moisture, the sensor measures the change and the pump is switched off when the set point is reached – the output affects the input, so the loop is closed ✓. (4)

2.3 Hysteresis is the difference between the switch-on point (30%) and the switch-off point (60%) ✓✓. Without it the pump would switch on and off very quickly around one value ✓, which wears out the pump, relays and contacts and wastes energy ✓. (4)

2.4 Pump rung: I0.0, I0.1 and I0.2 in series ✓✓ driving Q0.0 ✓; fan rung: I0.0 and I0.3 in series ✓ driving Q0.1 ✓; lamp rung: NC I0.2 ✓ driving Q0.2 ✓; correct ladder symbols and addresses ✓, e.g. (8)

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|──[ I0.0 ]──[ I0.1 ]──[ I0.2 ]──────────( Q0.0 )──|   pump: AUTO, soil dry, water in tank
|──[ I0.0 ]──[ I0.3 ]──────────────────────( Q0.1 )──|   fan: AUTO and too hot
|──[/ I0.2 ]──────────────────────────────( Q0.2 )──|   tank low warning
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2.5 Any THREE ✓✓✓: does not rust or corrode; light and easy to cut and join; cheap; smooth inside, so little friction loss; resistant to fertilisers and chemicals; flexible (polyethylene) so it can be laid along the beds. (3)

[25]

QUESTION 3

3.1 A WBS breaks the whole project into smaller, manageable tasks and work packages, shown as a tree ✓✓. Drawing: project at the top ✓; main branches (e.g. design, procurement, installation, testing) ✓✓; at least SIX sensible tasks under the branches ✓✓ (e.g. survey site, design system, order parts, install tank and pipes, install solar system, wire controller, program, commission, train users); neat tree layout ✓ (8)

3.2.1 $\mathrm{E} = (60 \times 3) + (40 \times 5) + (5 \times 24)$ ✓ $= 180 + 200 + 120$ ✓ $= 500\ \mathrm{Wh}$ ✓ (3)

3.2.2 $\mathrm{P} = 500 \div 5 = 100\ \mathrm{W}$ ✓; with losses $100 \times 1{,}25$ ✓ $= 125\ \mathrm{W}$ ✓ (3)

3.2.3 $\mathrm{Capacity} = 500 \div 12$ ✓ $= 41{,}67\ \mathrm{Ah}$ ✓ (2)

3.3 The budget is the planned (estimated) cost before the project starts ✓; the actual cost is what was really spent ✓. Tracking the difference shows early if the project is overspending ✓, so the manager can act (cut costs, ask for more money) before the money runs out ✓. (4)

3.4 3.4.1 Shows how the components are connected electrically and how the circuit works ✓, used to understand and check the circuit and find faults ✓. 3.4.2 Shows the real positions of terminals, wire numbers and colours ✓, used by the electrician to connect each wire correctly ✓. 3.4.3 Shows where the tank, pump, pipes, sensors, fan and panels are placed with dimensions ✓, used to mark out and install the parts in the right places ✓. (6)

3.5 Any FOUR items, each with a reason ✓✓✓✓, e.g. safety boots – protect the feet from falling objects; gloves – protect the hands from cuts and hot surfaces; safety glasses – protect the eyes when drilling and cutting; hard hat – protects the head when working under the roof; sun hat and sunscreen – protect against sunburn on the roof; insulated gloves – protect against electric shock. (4)

[30]

QUESTION 4

4.1 Measure the sensor reading in dry soil and in fully wet soil ✓; set these as 0% and 100% in the controller (or note the values) ✓; check the reading against a meter or the feel of the soil and adjust the set points (30% and 60%) until the pump switches at the right moisture ✓. (3)

4.2 Centre labelled with the system ✓; any FIVE legs with sensible tests ✓✓✓✓ (−1 if fewer than five), e.g. pump switches on when dry and off when wet; fan switches on above the set temperature; pump stops when the tank is low and the lamp comes on; solar panel charges the battery; data logger records correctly; no water leaks; emergency switch-off works (5)

4.3 A list of small faults or unfinished work found during the final inspection ✓ that the contractor must fix before the project is signed off ✓. (2)

4.4 Any THREE ✓✓✓: air in the pump (not primed); blocked filter, pipe or drippers; closed or faulty valve; a burst or disconnected pipe; pump running in the wrong direction or worn impeller; suction pipe above the water level. (3)

4.5 They must know how to use, check and switch off the system safely ✓, and how to do simple maintenance and recognise faults so that the plants are protected when the installer is not there ✓. (2)

[15]

QUESTION 5

5.1 Suitable headings (task, how often, done by / signed) ✓; six sensible tasks with the correct frequency ✓✓✓✓✓, e.g. weekly – check the water level and filters, check that the logger is recording; monthly – clean the solar panels, test the float switch and sensors; yearly – service the pump, check all wiring and battery capacity. (6)

5.2 Isolate the system or switch off at the charge controller ✓; clean early in the morning or late afternoon with clean water and a soft brush, without harsh chemicals or high pressure ✓; check for cracks, loose mountings, damaged cables and corrosion, and use a safe ladder or work from the ground ✓. (3)

5.3 It records values such as soil moisture, temperature, pump and fan running times, battery voltage and water level, with the date and time ✓✓. The records show trends and unusual readings, e.g. a pump running longer than normal (blocked pipe) or a battery that no longer charges, so faults can be fixed before the plants die ✓✓. (4)

5.4 A short circuit is a low-resistance connection between conductors (e.g. live and neutral, or + and −) that causes a very large current ✓✓. An earth fault is a connection between a live conductor and earth or a metal part that should not carry current, which can cause electric shock ✓✓. (4)

5.5 Any THREE ✓✓✓: show what work was done and when; help to plan future maintenance; show repeated faults; prove that safety checks were done; help with warranty claims; help to estimate costs and order spares. (3)

[20]

TOTAL: 100

END OF MARKING GUIDELINE